

UCL DEPARTMENT OF COMPUTER SCIENCE

COMP0080: Graphical Models

Topic-Grouped Past Examination Questions & Solutions

Categorized from 2021–2022, 2023–2024, and Classic Sample Examination Papers

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Master Course Syllabus & Topic Roadmap

This revision guide categorizes ****all past examination questions**** for UCL COMP0080 (Graphical Models) by topic, explicitly tagging the source examination year for every question:

- **2021–22 Final Examination** (COMP0080)
- **2023–24 Final Examination** (COMP0080A7PF / COMP0080A7UF)
- **Classic Sample Examination Paper** (COMPGI08 / COMPM056)

Topic Area	Exam Questions & Core Algorithms Covered
Topic 1: Representations	Independence in Joint Tables, Pairwise Independence, Markov Net Separation, State Probabilities, Markov Equivalence, Verma & Pearl Algorithm.
Topic 2: Exact Inference	Factor Graphs, Sum-Product / Belief Propagation, Junction Tree Algorithm (JTA), Moralisation, Triangulation, Running Intersection Property.
Topic 3: Time Series	Hidden Markov Models (HMMs), Forward Filtering (α), Backward Smoothing (β), Likelihood, Prediction, Viterbi Decoding, Exact Sampling.
Topic 4: Learning	Maximum Likelihood Estimation (MLE), Bipartite Markov Nets, Empirical KL Minimization, Randomized Response Surveys, Baum-Welch EM.
Topic 5: Approximate Inference	Matrix KL Divergence, Variational Free Energy ($F(q)$), Mode-Seeking vs Mode-Covering KL, Rejection Sampling, MCMC, Gibbs & MH.

TOPIC 1: Representations & Independence in Graphical Models

Question 1.1: Independence in Joint Probability Tables

[Source: 2021–22 Final Exam, Question 1(a)]

Suppose that X_1 and X_2 are two random variables, where each can take one of three states. The joint distribution $P(X_1 = i, X_2 = j)$ is described by matrix P :

$$P = \begin{pmatrix} 0.06 & 0.37 & 0.12 \\ 0.01 & 0.07 & 0.02 \\ 0.03 & 0.26 & 0.06 \end{pmatrix}$$

1. Are X_1 and X_2 independent? Support your answer. [2 Marks]
2. What is the minimal number of entries in the table that you need to change to make them independent while keeping the table a valid probability distribution (non-negative and summing to 1.0)? State the changed entries and the new matrix. [4 Marks]

Solution 1.1

1. **Testing Independence:** Check row marginal sums r_i and column marginal sums c_j :

$$r = [0.55, 0.10, 0.35]^T, \quad c = [0.10, 0.70, 0.20]^T$$

Testing entry $(1, 1)$: $r_1 c_1 = 0.55 \times 0.10 = 0.055 \neq 0.06 = P_{11}$. Since $P_{11} \neq r_1 c_1$, the variables are **not independent**.

2. **Minimal Entries Modification Proof:** An independent probability matrix is rank 1 ($P = rc^T$). All 2×2 subdeterminants must equal zero. Subdeterminants $\det(P_{1:2,1:2}) = 0.0005 \neq 0$ and $\det(P_{2:3,2:3}) = -0.0010 \neq 0$ share only entry P_{22} . Setting $P_{22} = x$ yields conflicting requirements ($x = 0.06167$ vs $x = 0.08667$). Thus ****changing 1 entry is impossible****.

Minimal changes required: ****2 entries****. Inspecting columns 1 and 3: row ratio is $6 : 1 : 3$. Adjusting Column 2 to match ratio $6 : 1 : 3$ with reference $P_{22} = 0.07$: - $P_{12} = 6 \times 0.07 = 0.42$ (changed from 0.37) - $P_{32} = 3 \times 0.07 = 0.21$ (changed from 0.26)

$$P_{\text{new}} = \begin{pmatrix} 0.06 & 0.42 & 0.12 \\ 0.01 & 0.07 & 0.02 \\ 0.03 & 0.21 & 0.06 \end{pmatrix} = \begin{pmatrix} 0.60 \\ 0.10 \\ 0.30 \end{pmatrix} \begin{pmatrix} 0.10 & 0.70 & 0.20 \end{pmatrix}$$

Question 1.2: Dependencies and Pairwise Independence (XOR)

[Source: 2021–22 Final Exam, Q1(b-c) & 2023–24 Final Exam, Q1(a)]

1. Suppose $A \not\perp B$ and $B \not\perp C$. Does that necessarily imply $A \not\perp C$? Prove or provide a counterexample. [3 Marks]
2. Give an example of a joint distribution in which $A \perp B$, $B \perp C$, $C \perp A$ (pairwise independence), but $A \not\perp B \mid C$. [3 Marks]

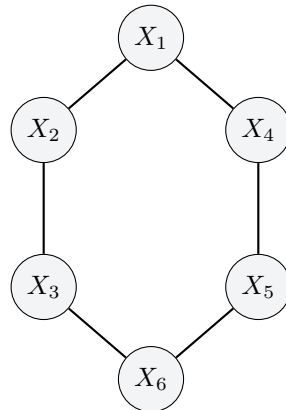
Solution 1.2

1. **Dependency Transitivity Counterexample:** Let $A, C \sim \text{Bernoulli}(0.5)$ be independent coin tosses ($A \perp C$). Define $B = (A, C)$. Since B contains A , $A \not\perp B$. Since B contains C , $B \not\perp C$. Yet $A \perp C$ by construction. Thus dependency is ****not transitive****.
2. **Pairwise Independence with Conditional Dependence (XOR):** Let $A, B \in \{0, 1\}$ be independent fair coins ($P(A) = P(B) = 0.5$). Let $C = A \oplus B$ (XOR). - $P(A, B) = P(A)P(B) = 0.25 \implies A \perp B$. - $P(B, C) = P(B)P(C) = 0.25 \implies B \perp C$. - $P(A, C) = P(A)P(C) = 0.25 \implies A \perp C$. - However, given $C = 0 \implies A = B$, so $P(A = 1, B = 1 \mid C = 0) = 0.5 \neq P(A = 1 \mid C = 0)P(B = 1 \mid C = 0) = 0.25$. They are ****conditionally dependent given C****.

Question 1.3: Markov Network Marginal Independence

[Source: 2021–22 Final Exam, Q1(d) & 2023–24 Final Exam, Q1(c)]

Consider the undirected Markov Network on variables $X_1, \dots, X_6$:



Write down the set of all conditional and marginal independence statements guaranteed to hold for the marginal distribution $P(X_2, X_4, X_6)$. Justify using graph separation. [6 Marks]

Solution 1.3

Marginalizing out X_1, X_3, X_5 connects their immediate neighbors: - Eliminating X_1 (connected to X_2, X_4) adds edge (X_2, X_4) . - Eliminating X_3 (connected to X_2, X_6) adds edge (X_2, X_6) . - Eliminating X_5 (connected to X_4, X_6) adds edge (X_4, X_6) . The marginal graph on $\{X_2, X_4, X_6\}$ forms a fully connected 3-clique K_3 . Since every pair is directly connected, no graph separation exists. Guaranteed independence set: $\emptyset$ (empty set).

Question 1.4: Exact State Probability in Pairwise Markov Net

[Source: 2023–24 Final Exam, Question 1(b)]

Consider a pairwise Markov Network on 3 binary variables $X_1, X_2, X_3 \in \{0, 1\}$ with joint distribution $P(X) \propto \Phi(X_1, X_2)\Phi(X_2, X_3)\Phi(X_3, X_1)$ where:

$$\Phi(X_i, X_j) = \begin{cases} 2 & \text{if } X_i = X_j \\ 1 & \text{if } X_i \neq X_j \end{cases}$$

Compute the exact probability $P(X_1 = X_2 = X_3)$ of all three variables being in the same state. [5 Marks]

Solution 1.4

Total state space size is $2^3 = 8$ states:

- Equal States $(0, 0, 0)$ and $(1, 1, 1)$ (2 states): $\Phi = 2 \times 2 \times 2 = 8$. Weight sum $= 2 \times 8 = 16$.
- Unequal States (6 states): 2 edges agree ($\Phi = 2$), 1 edge disagrees ($\Phi = 1$). Weight $\Phi = 2 \times 1 \times 1 = 2$. Weight sum $= 6 \times 2 = 12$.

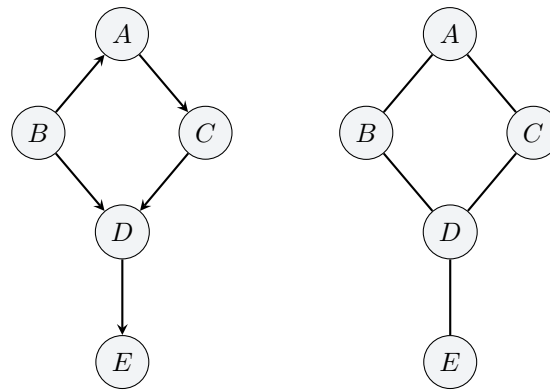
Partition function: $Z = 16 + 12 = 28$.

$$P(X_1 = X_2 = X_3) = \frac{16}{28} = \frac{4}{7} \approx 0.5714$$

Question 1.5: Markov Equivalence & Graph Comparisons

[Source: 2021–22 Final Exam, Q1(e) & 2023–24 Final Exam, Q1(d)]

1. Compare Model 1 (DAG) and Model 2 (Markov Network):



Model 1 (DAG) Model 2 (Markov Network)

Do they encode the same set of independence assumptions? List all differences. [6 Marks]

- Compare G1 ($A \rightarrow C \leftarrow B$), G2 ($A \leftarrow C \rightarrow B$), G3 ($A \rightarrow C \rightarrow B$), G4 ($A \leftarrow C \leftarrow B$). Which are Markov equivalent? [6 Marks]

Solution 1.5

- Skeleton is identical: $\{(A, B), (A, C), (B, D), (C, D), (D, E)\}$. Model 1 has immorality $B \rightarrow D \leftarrow C$. - $B \perp\!\!\!\perp C \mid A$: **True** in Model 1, **False** in Model 2. - $B \perp\!\!\!\perp C \mid \{A, D\}$: **False** in Model 1, **True** in Model 2. Not Markov equivalent.
- G1** (Collider): Immorality at $C \implies A \perp\!\!\!\perp B$. **G2, G3, G4**: No immoralities, same skeleton $\implies A \perp\!\!\!\perp B \mid C$. **G2, G3, G4 are Markov equivalent**.

Question 1.6: Fundamental Definitions & Universal Factorizations

[Source: Classic Sample Exam, Question 1(a-b)]

- Explain what dependency between discrete random variables X_i and X_j means mathematically. [2 Marks]
- Define a Belief Network. Why must its graphical structure be a Directed Acyclic Graph (DAG)? [2 Marks]
- Prove why any joint distribution $p(x_1, \dots, x_N)$ can be written as a Belief Network. [2 Marks]
- Explain why the lack of an edge in a DAG implies an independence statement, whereas the presence of an edge does not guarantee dependence. [3 Marks]

Solution 1.6

- X_i and X_j are dependent if $p(x_i, x_j) \neq p(x_i)p(x_j)$.
- Factorization $p(x_{1:N}) = \prod_i p(x_i \mid \text{pa}(x_i))$. Must be a DAG to prevent cyclic circular probability definitions ($p(a|b)p(b|a)$).
- By chain rule: $p(x_{1:N}) = p(x_1)p(x_2|x_1)p(x_3|x_1, x_2) \dots p(x_N|x_{1:N-1})$.
- Missing edge guarantees conditional independence. Edge presence allows dependence but does not force it (e.g. CPT can be uniform).

Question 1.7: DAG to Undirected Conversion Counterexample

[Source: Classic Sample Exam, Question 1(b.iii)]

Let DAG A have adjacency matrix A_{ij} . Form undirected adjacency matrix $B_{ij} = A_{ij} \vee A_{ji}$. Can every Belief Network with DAG A be represented by a Markov Network with adjacency B ? Explain or give counterexample. [3 Marks]

Solution 1.7

Consider collider DAG $A \rightarrow B \leftarrow C$. Skeleton matrix $B = A \vee A^T$ has edges (A, B) and (B, C) , but no edge (A, C) . In undirected graph B , path $A - B - C$ means $A \not\perp C$. But in DAG A , $A \perp\!\!\!\perp C$ holds. Thus B cannot represent the DAG without adding moral edge (A, C) .

Question 1.8: Verma & Pearl Markov Equivalence Algorithm

[Source: *Classic Sample Exam, Question 1(c)*]

1. Define Markov Equivalence between two graphs. [2 Marks]
2. State an algorithm to determine if two DAGs are Markov equivalent. [2 Marks]
3. Check equivalence for Pair 1: $(a \rightarrow b \rightarrow d, a \rightarrow c \rightarrow d)$ vs $(a \leftarrow b \rightarrow d, a \rightarrow c \rightarrow d)$. [3 Marks]
4. Check equivalence for Pair 2: $(a \rightarrow b \rightarrow d, a \rightarrow c \rightarrow d)$ vs $(a \rightarrow b \rightarrow d, a \rightarrow c \leftarrow d)$. [3 Marks]

Solution 1.8

1. Two DAGs are Markov equivalent if they encode identical conditional independence sets.
2. **Algorithm (Verma & Pearl):** Two DAGs are Markov equivalent if and only if they have the **same skeleton** and the **same immoralities** (unmarried colliders).
3. Pair 1: Both have skeleton $\{(a, b), (a, c), (b, d), (c, d)\}$ and no immoralities. **Markov equivalent**.
4. Pair 2: Graph 1 has no immoralities. Graph 2 has immorality $b \rightarrow d \leftarrow c$. **Not Markov equivalent**.

TOPIC 2: Exact Inference & Factor Graphs

Question 2.1: The Junction Tree Algorithm

[Source: 2021–22 Final Exam, Q2.2 & 2023–24 Final Exam, Q2(b)]

Consider the joint distribution that factorizes as:

$$p(a, b, c, d, e, f) = p(a)p(b | a)p(c | b) \\ \times p(d | c)p(e | d)p(f | a, e)$$

1. Draw the DAG representing this distribution. [3 Marks]
2. Draw the moralised version of this graph. Explain parent marrying. [3 Marks]
3. Draw the triangulated version of the graph with the smallest possible maximum clique size. [3 Marks]
4. Construct a Junction Tree and verify the Running Intersection Property (RIP). [3 Marks]
5. Describe potential initialization, absorption, and message schedule. [3 Marks]

Solution 2.1

1. **DAG:** Edges $a \rightarrow b \rightarrow c \rightarrow d \rightarrow e$, $a \rightarrow f$, $e \rightarrow f$.
2. **Moralisation:** Marry parents of f (a and e) with moral edge $a - e$. Convert edges to undirected.
3. **Triangulation:** 5-cycle $a - b - c - d - e - a$ triangulated with chords $a - c$ and $a - d$. Maximal cliques $C_1 = \{a, b, c\}$, $C_2 = \{a, c, d\}$, $C_3 = \{a, d, e\}$, $C_4 = \{a, e, f\}$ (all size 3).
4. **Junction Tree & RIP:** Chain $C_1 \overset{\{a,c\}}{-} C_2 \overset{\{a,d\}}{-} C_3 \overset{\{a,e\}}{-} C_4$. Variable a forms a continuous sub-tree $C_1 - C_2 - C_3 - C_4$. RIP holds.
5. **Absorption Schedule:** Forward pass $C_1 \rightarrow C_2 \rightarrow C_3 \rightarrow C_4$, Backward pass $C_4 \rightarrow C_3 \rightarrow C_2 \rightarrow C_1$. Update equation $\psi(V)^{\text{new}} = \psi(V)^{\frac{\phi(S)^{\text{new}}}{\phi(S)^{\text{old}}}}$.

Question 2.2: Sum-Product Algorithm on Factor Graphs

[Source: Classic Sample Exam, Question 2(a)]

Describe the sum-product algorithm for computing marginals in a factor graph. Explain which factor graph topologies the algorithm is applicable to. [4 Marks]

Solution 2.2

The sum-product algorithm performs exact marginal inference on tree-structured factor graphs by passing messages between variable nodes i and factor nodes f :

$$\mu_{f \rightarrow i}(x_i) = \sum_{\chi_f \setminus \{i\}} f(x_i, \chi_f) \prod_{j \in \text{ne}(f) \setminus \{i\}} \mu_{j \rightarrow f}(x_j) \\ \mu_{i \rightarrow f}(x_i) = \prod_{g \in \text{ne}(i) \setminus \{f\}} \mu_{g \rightarrow i}(x_i)$$

Applicable for exact marginals on single-tree factor graphs (no cycles).

TOPIC 3: Time Series & Hidden Markov Models (HMMs)

Question 3.1: HMM Posterior Graph & Exact Sampling

[Source: 2021–22 Final Exam, Q2(b.i-ii) & Classic Sample Exam, Q2(b.vii)]

1. When sequence $v_{1:T}$ is observed, it induces posterior distribution $p_v(h_{1:T}) = p(h_{1:T} \mid v_{1:T})$. What is the graphical model of this posterior? Is $h_1 \perp\!\!\!\perp h_T$? [3 Marks]
2. Can we efficiently sample from $p_v(h_{1:T})$ exactly? Describe how. [3 Marks]

Solution 3.1

1. Linear chain graph $h_1 - h_2 - \dots - h_T$. h_1 and h_T are ****dependent****.
2. Yes, via Forward-Filtering Backward-Sampling in $O(T)$ time:

$$p(h_{1:T} \mid v_{1:T}) = p(h_T \mid v_{1:T}) \prod_{t=1}^{T-1} p(h_t \mid h_{t+1}, v_{1:t})$$

Sample $h_T^* \sim \alpha_T(h_T)$, then sample backwards $h_t^* \sim p(h_t \mid h_{t+1}^*, v_{1:t}) \propto p(h_{t+1}^* \mid h_t) \alpha_t(h_t)$.

Question 3.2: Deterministic State Decoupling & Step-by-Step Filtering

[Source: 2021–22 Final Exam, Question 2.3]

Given $v_t \in \{0, 1, 2\}$, $h_t \in \{0, 1\}$, $p(h_1 = 1) = 0.5$, transition matrix $T = \begin{pmatrix} 0.5 & 0.8 \\ 0.5 & 0.2 \end{pmatrix}$, emission matrix

$B = \begin{pmatrix} 0.3 & 0.5 \\ 0.6 & 0 \\ 0.1 & 0.5 \end{pmatrix}$, and sequence $v_{1:10} = [0, 1, 0, 2, 0, 2, 1, 0, 2, 0]$:

1. Is $h_1 \perp\!\!\!\perp h_{10} \mid v_{1:10}$ for this specific sequence? Why? [2 Marks]
2. Compute exact filtering distributions $p(h_8 \mid v_{1:8})$ and $p(h_9 \mid v_{1:9})$. [3 Marks]

Solution 3.2

1. Emission row 1 for state $h = 1$ is $B_{1,1} = p(v = 1 \mid h = 1) = 0$. Since $v_2 = 1$ and $v_7 = 1$, states are pinned deterministically: $h_2 = 0$ and $h_7 = 0$ with probability 1. Conditioning on $h_7 = 0$ d-separates h_1 from h_{10} . They are ****independent****.

2. Filtering calculations: - Base at $t = 7$: $p(h_7 \mid v_{1:7}) = \begin{pmatrix} 1.0 \\ 0.0 \end{pmatrix}$. - Predict $t = 8$: $p(h_8 \mid v_{1:7}) = Tp(h_7 \mid v_{1:7}) = \begin{pmatrix} 0.5 & 0.8 \\ 0.5 & 0.2 \end{pmatrix} \begin{pmatrix} 1.0 \\ 0.0 \end{pmatrix} = \begin{pmatrix} 0.5 \\ 0.5 \end{pmatrix}$. - Observe $v_8 = 0$: Unnormalized $\begin{pmatrix} 0.3 \times 0.5 \\ 0.5 \times 0.5 \end{pmatrix} = \begin{pmatrix} 0.15 \\ 0.25 \end{pmatrix} \Rightarrow p(h_8 \mid v_{1:8}) = \begin{pmatrix} 0.375 \\ 0.625 \end{pmatrix} = \begin{pmatrix} 3/8 \\ 5/8 \end{pmatrix}$. - Predict $t = 9$: $p(h_9 \mid v_{1:8}) = Tp(h_8 \mid v_{1:8}) = \begin{pmatrix} 0.5 & 0.8 \\ 0.5 & 0.2 \end{pmatrix} \begin{pmatrix} 0.375 \\ 0.625 \end{pmatrix} = \begin{pmatrix} 0.6875 \\ 0.3125 \end{pmatrix}$. - Observe $v_9 = 2$: Unnormalized $\begin{pmatrix} 0.1 \times 0.6875 \\ 0.5 \times 0.3125 \end{pmatrix} = \begin{pmatrix} 0.06875 \\ 0.15625 \end{pmatrix} \Rightarrow p(h_9 \mid v_{1:9}) = \begin{pmatrix} 11/36 \\ 25/36 \end{pmatrix} \approx \begin{pmatrix} 0.3056 \\ 0.6944 \end{pmatrix}$.

Question 3.3: HMM Filtering, Smoothing & Viterbi Recursions

[Source: Classic Sample Exam, Question 2(b.i-vi)]

1. Derive forward filtering $\alpha_t(h_t) = p(h_t, v_{1:t})$. [3 Marks]
2. Derive backward smoothing $\beta_t(h_t) = p(v_{t+1:T} \mid h_t)$. [3 Marks]
3. Compute sequence likelihood $p(v_{1:T})$. [5 Marks]
4. Derive prediction $p(v_{T+1} \mid v_{1:T})$. [5 Marks]
5. Derive Viterbi algorithm for $\operatorname{argmax}_{h_{1:T}} p(h_{1:T} \mid v_{1:T})$. [3 Marks]

Solution 3.3

1. $\alpha_t(h_t) = p(v_t \mid h_t) \sum_{h_{t-1}} p(h_t \mid h_{t-1}) \alpha_{t-1}(h_{t-1})$.
2. $\beta_t(h_t) = \sum_{h_{t+1}} p(v_{t+1} \mid h_{t+1}) p(h_{t+1} \mid h_t) \beta_{t+1}(h_{t+1})$.
3. $p(v_{1:T}) = \sum_{h_T} \alpha_T(h_T)$.
4. $p(v_{T+1} \mid v_{1:T}) = \sum_{h_{T+1}} p(v_{T+1} \mid h_{T+1}) \sum_{h_T} p(h_{T+1} \mid h_T) p(h_T \mid v_{1:T})$.
5. $V_t(h_t) = p(v_t \mid h_t) \max_{h_{t-1}} [p(h_t \mid h_{t-1}) V_{t-1}(h_{t-1})]$.

TOPIC 4: Parameter Learning & EM Algorithm

Question 4.1: Bipartite Disease-Symptom Markov Network

[Source: 2021–22 Final Exam, Question 2.1]

Given $p(s, d) = \frac{1}{Z} \exp(s^T W d + a^T s + b^T d)$:

1. Draw the Markov Network. Describe structure. [3 Marks]
2. Derive $p(s | d)$. Is it factorized? [4 Marks]
3. Derive log-likelihood L for N records (s^n, d^n) . [3 Marks]
4. Derive gradients $\frac{\partial L}{\partial W_{ij}}, \frac{\partial L}{\partial a_i}, \frac{\partial L}{\partial b_j}$. [5 Marks]
5. Efficiency in general case? [3 Marks]

Solution 4.1

1. Bipartite graph between s_i and d_j .
2. $p(s | d) = \prod_{i=1}^S \sigma(W_i \cdot d + a_i)^{s_i} (1 - \sigma(W_i \cdot d + a_i))^{1-s_i}$. Fully factorized.
3. $L = \sum_{n=1}^N [(s^n)^T W d^n + a^T s^n + b^T d^n] - N \log Z(W, a, b)$.
4. $\frac{\partial L}{\partial W_{ij}} = \sum_{n=1}^N (s_i^n d_j^n - \langle s_i d_j \rangle_{p(s,d)})$.
5. Infeasible in general case (2^{S+D} sum for partition function Z).

Question 4.2: Empirical KL Minimization & Edge Removal

[Source: 2023–24 Final Exam, Question 2.2.1]

Belief Net $A \rightarrow B, A \rightarrow C$. Learning parameters by Maximum Likelihood minimizes $KL(Q \| P)$ where Q is empirical distribution and P is model distribution.

1. Optimal setting of distribution P and value of $KL(Q \| P)$? [5 Marks]
2. Remove edge $A \rightarrow B$: impact on optimal KL divergence? [5 Marks]
3. Dataset conditions for equal KL divergence? [5 Marks]

Solution 4.2

1. $P^*(A, B, C) = Q(A, B, C) \implies KL(Q \| P^*) = 0$.
2. Constrains P to factorizations $P(A)P(B)P(C|A)$. Minimum KL divergence **cannot decrease** ($KL(Q \| P^*) \geq 0$).
3. Holds iff A and B are **marginally independent** under empirical data $Q(A, B) = Q(A)Q(B)$.

Question 4.3: Randomized Response Survey Procedure & EM

[Source: 2021–22 Final Exam, Q2.4 & 2023–24 Final Exam, Q2(c)]

1. Graphical model & CPTs. [2 Marks]
2. Truthfulness posterior $P(C_1 = 1 | Y = 1)$ for $p_1 = 0.5$. [2 Marks]
3. MLE for $p_1 = 0.5$. [2 Marks]
4. EM updates for unknown (p_1, p_2) . [2 Marks]
5. Uniqueness for $D = (\text{yes}, \text{yes}, \text{yes}, \text{no})$? [2 Marks]

Solution 4.3

1. $H \rightarrow Y \leftarrow C_1$. $P(Y = 1 \mid H, C_1) = C_1 H + (1 - C_1) \cdot 0.5$.
2. $P(C_1 = 1 \mid Y = 1) = \frac{2p_2}{2p_2+1}$.
3. $\hat{p}_2 = \max(0, \min(1, 2\frac{n_y}{n} - 0.5))$.
4. **E-step:** $\langle C_{1,i} \rangle$ and $\langle H_i \rangle$ posteriors. **M-step:** $p_1^{\text{new}} = \frac{1}{n} \sum_i \langle C_{1,i} \rangle$, $p_2^{\text{new}} = \frac{1}{n} \sum_i \langle H_i \rangle$.
5. $\hat{\theta} = 0.75 \implies p_1(p_2 - 0.5) = 0.25$. 1D solution manifold. **Not unique**.

Question 4.4: Baum-Welch EM Derivation & Monotonicity Proof

[Source: Classic Sample Exam, Question 2(c)]

1. Derive Baum-Welch E and M steps for transition A and emission B . [6 Marks]
2. Prove log-likelihood is monotonically non-decreasing using free energy bound. [3 Marks]

Solution 4.4

1. **E-step:** $\gamma_t^n(i) = \frac{\alpha_t^n(i)\beta_t^n(i)}{p(v_{1:T}^n)}$ and $\xi_t^n(j, i) = \frac{\alpha_{t-1}^n(j)A_{ij}B_{v_t^n, i}\beta_t^n(i)}{p(v_{1:T}^n)}$. **M-step:** $\hat{A}_{ij} = \frac{\sum_n \sum_t \xi_t^n(j, i)}{\sum_n \sum_t \gamma_{t-1}^n(j)}$
and $\hat{B}_{ki} = \frac{\sum_n \sum_{t: v_t=k} \gamma_t^n(i)}{\sum_n \sum_t \gamma_t^n(i)}$.
2. Log-likelihood decomposition $\log p(V \mid \theta) = \mathcal{B}(q, \theta) + KL(q \parallel p)$. E-step sets $KL = 0$, M-step maximizes $\mathcal{B}(q, \theta)$, ensuring $\log p(V \mid \theta^{\text{new}}) \geq \log p(V \mid \theta^{\text{old}})$.

TOPIC 5: Approximate Inference & Sampling

Question 5.1: Matrix KL Divergence Computation

[Source: 2021–22 Final Exam, Question 3.1]

Given $P(X) = (0.25, 0.25, 0.25, 0.25)^T$, $Q(X) = (0.5, 0.25, 0.125, 0.125)^T$, and matrices P, Q :

$$P = \begin{pmatrix} 0.25 & 0.5 & 0 & 0.2 \\ 0.25 & 0 & 0.5 & 0.3 \\ 0.25 & 0 & 0 & 0.1 \\ 0.25 & 0.5 & 0.5 & 0.4 \end{pmatrix}, \quad Q = \begin{pmatrix} 0.1 & 0 & 0 & 1 \\ 0 & 0 & 0.5 & 0 \\ 0.9 & 0.5 & 0 & 0 \\ 0 & 0.5 & 0.5 & 0 \end{pmatrix}$$

Compute marginals $P(Y), Q(Y)$, and divergences $KL(P(Y) \parallel Q(Y))$ and $KL(Q(Y) \parallel P(Y))$. [5 Marks]

Solution 5.1

$$P(Y) = [0.2375, 0.2625, 0.0875, 0.4125]^T, \quad Q(Y) = [0.1750, 0.0625, 0.5750, 0.1875]^T.$$

$$\begin{aligned} KL(P \parallel Q) &= 0.2375 \ln \left(\frac{0.2375}{0.1750} \right) + 0.2625 \ln \left(\frac{0.2625}{0.0625} \right) + 0.0875 \ln \left(\frac{0.0875}{0.5750} \right) + 0.4125 \ln \left(\frac{0.4125}{0.1875} \right) \\ &= \mathbf{0.6097} \text{ nats} \quad (0.8797 \text{ bits}) \end{aligned}$$

$$\begin{aligned} KL(Q \parallel P) &= 0.1750 \ln \left(\frac{0.1750}{0.2375} \right) + 0.0625 \ln \left(\frac{0.0625}{0.2625} \right) + 0.5750 \ln \left(\frac{0.5750}{0.0875} \right) + 0.1875 \ln \left(\frac{0.1875}{0.4125} \right) \\ &= \mathbf{0.7916} \text{ nats} \quad (1.1420 \text{ bits}) \end{aligned}$$

Question 5.2: Mean-Field Variational Free Energy Derivation

[Source: 2021–22 Final Exam, Q3.2 & 2023–24 Final Exam, Q3(a)]

For 3-node pairwise Markov network $P(X) \propto \prod_{i \sim j} \Phi(X_i, X_j)$ where $\Phi(X_i, X_j) = 2$ if $X_i = X_j$ and 1 if $X_i \neq X_j$, derive explicit free energy $F(q)$ under fully factorized $q(X) = \prod_{i=1}^3 q_i(X_i)$. [5 Marks]

Solution 5.2

$$F(q) = \sum_{i=1}^3 [\mu_i \log \mu_i + (1 - \mu_i) \log(1 - \mu_i)] - \log 2 \sum_{(i,j)} [\mu_i \mu_j + (1 - \mu_i)(1 - \mu_j)]$$

Question 5.3: Mode-Seeking vs Mode-Covering KL Divergence

[Source: Classic Sample Exam, Question 3(a)]

Explain the operational difference between minimizing $KL(P \parallel Q)$ vs $KL(Q \parallel P)$ when approximating bimodal target P with unimodal proposal Q . [5 Marks]

Solution 5.3

- **Mode-Seeking ($KL(Q \parallel P)$):** Zero-forcing. Q contracts to cover a single peak of P . - **Mode-Covering ($KL(P \parallel Q)$):** Zero-avoiding. Q spreads wide to cover all modes of P .

Question 5.4: Rejection Sampling on Lattices

[Source: 2021–22 Final Exam, Q3(b) & 2023–24 Final Exam, Q3(b)]

Compute exact average acceptance rates $P(\text{accept}) = Z/2^{MN}$ for: - $M = 2, N = 1$: $Z = 2 + 2e^{-1} \approx 2.7358 \implies P(\text{accept}) = \mathbf{0.6839}$ (68.39%). - $M = 2, N = 2$: $Z = 2 + 12e^{-2} + 2e^{-4} \approx 3.6606 \implies P(\text{accept}) = \mathbf{0.2288}$ (22.88%).

Question 5.5: MCMC, Gibbs & Metropolis-Hastings

[Source: 2021–22 Final Exam, Q3(c), 2023–24 Final Exam, Q3(c) & Classic Sample Exam, Q3(b)]

Detailed balance is sufficient, not necessary. Counterexample: Lazy cyclic 3-state chain $T = \begin{pmatrix} 0.5 & 0.5 & 0 \\ 0 & 0.5 & 0.5 \\ 0.5 & 0 & 0.5 \end{pmatrix}$

with $P = [1/3, 1/3, 1/3]^T$. $PT = P$ holds, but $P(1)T(2|1) = 1/6 \neq 0 = P(2)T(1|2)$. Detailed balance is violated.